$$\mathbf{n}(t) = \mathbf{L}^t \mathbf{n}(0)$$

$$\overbrace{\begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_m(t) \end{pmatrix}}^{\mathbf{n}(t)} = \overbrace{\begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-1} b_m & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & s_{m-1} & 0 \end{pmatrix}}^{\mathbf{L}^t} \overbrace{\begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_m(0) \end{pmatrix}}^{\mathbf{n}(0)}$$

Mathematics

$\mathbf{n}(t) = \mathbf{L}^t \mathbf{n}(0)$

$$\overbrace{\begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_m(t) \end{pmatrix}}^{\mathbf{n}(t)} = \overbrace{\begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-1} b_m & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & s_{m-1} & 0 \end{pmatrix}}^{\mathbf{L}^t} \overbrace{\begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_m(0) \end{pmatrix}}^{\mathbf{n}(0)}$$

Algebra

Matrix,
Linear transformation,
Eigen vector,
Geometry Eigenvalue

Part 4-4

Matrix and vector multiplication

$$\mathbf{Ax} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Method 1: $\mathbf{a}_1 = \begin{pmatrix} a_{11} & a_{12} \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} a_{21} & a_{22} \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} -\mathbf{a}_1- \\ -\mathbf{a}_2- \end{pmatrix}$

$$\mathbf{Ax} = \begin{pmatrix} -\mathbf{a}_1- \\ -\mathbf{a}_2- \end{pmatrix} \begin{pmatrix} | \\ \mathbf{x} \\ | \end{pmatrix}$$

$$= \begin{pmatrix} \begin{pmatrix} a_{11} & a_{12} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{pmatrix}$$

Matrix and vector multiplication

$$\mathbf{Ax} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

Method 2: $\mathbf{a}_1 = \begin{pmatrix} a_{11} \\ a_{12} \end{pmatrix}$, $\mathbf{a}_2 = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$, $\mathbf{A} = \begin{pmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{pmatrix}$

$$\mathbf{Ax} = \begin{pmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= x\mathbf{a}_1 + y\mathbf{a}_2$$

$$= x \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} + y \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = \begin{pmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \mathbf{x}_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}, \quad \mathbf{x}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$\mathbf{e}_1 = \begin{pmatrix} a \\ c \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} b \\ d \end{pmatrix}$$

$$\mathbf{Ax}_0 = \overbrace{\begin{pmatrix} \mathbf{e}_1 & \mathbf{e}_2 \end{pmatrix}}^{\mathbf{A}} \overbrace{\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}}^{\mathbf{x}_0} = x_0 \mathbf{e}_1 + y_0 \mathbf{e}_2$$

$$= x_0 \begin{pmatrix} a \\ c \end{pmatrix} + y_0 \begin{pmatrix} b \\ d \end{pmatrix} = \begin{pmatrix} x_0 a + y_0 b \\ x_0 c + Y_0 d \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \mathbf{x}_1$$

$$\mathbf{Ax}_0 = \mathbf{x}_1$$

Linear transformation

Algebra

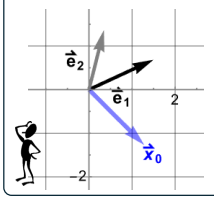
$$\mathbf{Ax}_0 = (\vec{\mathbf{e}}_1, \vec{\mathbf{e}}_2) \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

$$x_0 \begin{pmatrix} | \\ \vec{\mathbf{e}}_1 \\ | \end{pmatrix} + y_0 \begin{pmatrix} | \\ \vec{\mathbf{e}}_1 \\ | \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

Transformation

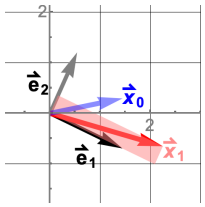
$$\vec{\mathbf{x}}_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \xrightarrow{\mathbf{A}} \vec{\mathbf{x}}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

Geometry



$\vec{e}_1 = \begin{pmatrix} a \\ c \end{pmatrix}$, $\vec{e}_2 = \begin{pmatrix} b \\ d \end{pmatrix}$, $\mathbf{A} = (\vec{e}_1, \vec{e}_2)$

$\begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \overset{\mathbf{A}}{\Longrightarrow} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$

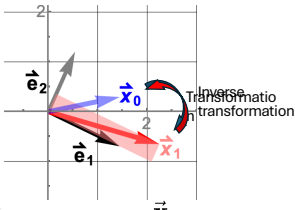


$\vec{e}_1$ and $\vec{e}_2$ are called **base vectors**

- ✓ Set base vectors in arbitrary directions.
- ✓ Change the direction and length of $\mathbf{x}_0$ as you like.
- ✓ Observe the directions and lengths of $\mathbf{x}_0$ and $\mathbf{x}_1$.

$$\mathbf{A}\,\mathbf{x}_0 = \mathbf{x}_1$$

Transformation $\quad \overbrace{\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}}^{\vec{x}_0} \overset{\mathbf{A}}{\Longrightarrow} \overbrace{\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}^{\vec{x}_1}$



Inverse transformation $\quad \overbrace{\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}}^{\vec{x}_0} \Longleftarrow \overbrace{\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}}^{\vec{x}_1}$

Inverse Matrix, Identity Matrix

transformation

$$\mathbf{x}' = \mathbf{A}\,\mathbf{x}$$

$$\Leftarrow \mathbf{A} \Leftarrow$$

$$\mathbf{x}' = \begin{pmatrix} x' \\ y' \end{pmatrix} \qquad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Longrightarrow \mathbf{A}^{-1} \Rightarrow$$

$$\mathbf{A}^{-1}\mathbf{x}' = \mathbf{x}$$

inverse transformatoin

$$\mathbf{A}^{-1}\underbrace{\mathbf{A}\mathbf{x}}_{\mathbf{x}'} = \underbrace{\mathbf{x}}_{\mathbf{x}} \qquad \underbrace{\mathbf{A}^{-1}\mathbf{A}}_{\mathbf{I}}\mathbf{x} = \mathbf{x} \qquad \mathbf{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Transformation

$\vec{e}_1 = \begin{pmatrix} a \\ c \end{pmatrix}$, $\vec{e}_2 = \begin{pmatrix} b \\ d \end{pmatrix}$, $\quad \mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (\vec{e}_1 \quad \vec{e}_2)$

$\vec{x}_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$, $\vec{x}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$

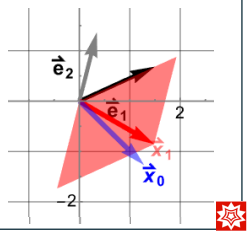
Algebra

$$\mathbf{A}\mathbf{x}_0 = (\vec{e}_1 \quad \vec{e}_2)\begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

$$x_0 \begin{pmatrix} | \\ \vec{e}_1 \\ | \end{pmatrix} + y_0 \begin{pmatrix} | \\ \vec{e}_2 \\ | \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

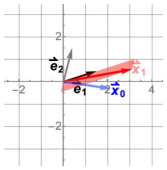
$$\vec{x}_0 = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} \overset{\mathbf{A}}{\Longrightarrow} \vec{x}_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

Geometry

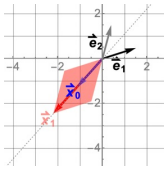


Geometry

$$\mathbf{A}\,\mathbf{x}_0 = \mathbf{x}_1$$



$$\mathbf{A}\overbrace{\mathbf{x}}^{\mathbf{x}_0} = \overbrace{\lambda\mathbf{x}}^{\mathbf{x}_1}$$



from German "its own"

- ✓ The special vectors are called eigen-vectors.
- ✓ The factors, λ, that makes $\mathbf{x}_0 = \lambda\,\mathbf{x}_1$ are called eigen-values.

Algebra

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

$$\mathbf{A}\mathbf{x} - \lambda\mathbf{x} = \mathbf{0}$$

$$\mathbf{A}\mathbf{x} - \lambda\mathbf{I}\mathbf{x} = \mathbf{0} \qquad \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$(\mathbf{A} - \lambda\mathbf{I})\,\mathbf{x} = \mathbf{0} \qquad \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x\begin{pmatrix} a-\lambda \\ c \end{pmatrix} + y\begin{pmatrix} b \\ d-\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

For the left-hand side and the right-hand side to be equal, x, y, and λ must satisfy certain

Algebra and Geometry

$$x\begin{pmatrix} a-\lambda \\ c \end{pmatrix} + y\begin{pmatrix} b \\ d-\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 3 & 0.8 \\ 1.5 & 3 \end{pmatrix}$, $\mathbf{u} = \begin{pmatrix} x \\ y \end{pmatrix}$, λ

$\mathbf{A}\cdot\mathbf{u} = \lambda\cdot\mathbf{u}$

$(\mathbf{A}-\lambda\cdot\mathbf{I})\cdot\mathbf{u} = \mathbf{0}$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} - \lambda\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

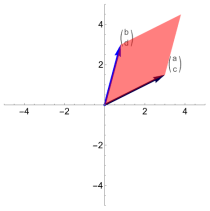
$\begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$x\begin{pmatrix} a-\lambda \\ c \end{pmatrix} + y\begin{pmatrix} b \\ d-\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$x\begin{pmatrix} 3-0 \\ 1.5 \end{pmatrix} + y\begin{pmatrix} 0.8 \\ 3-0 \end{pmatrix} = \begin{pmatrix} ? \\ ? \end{pmatrix}$

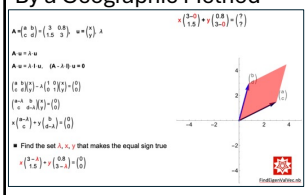
■ Find the set λ, x, y that makes the equal sign true

$x\begin{pmatrix} 3-\lambda \\ 1.5 \end{pmatrix} + y\begin{pmatrix} 0.8 \\ 3-\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$



FindEigenValVec.cc

By a Geographic Method



$$\mathbf{A} = \begin{pmatrix} 3 & 0.8 \\ 1.5 & 3 \end{pmatrix}$$

Eigenvalue	Eigen Vector
$\lambda_1 = 1.9,$	$\mathbf{u}_1 \propto \begin{pmatrix} -1 \\ 1.4 \end{pmatrix}$
$\lambda_2 = 4.1,$	$\mathbf{u}_2 \propto \begin{pmatrix} 1 \\ 1.4 \end{pmatrix}$



Algebra

Equation

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$$

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad x\begin{pmatrix} a-\lambda \\ c \end{pmatrix} + y\begin{pmatrix} b \\ d-\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & k \end{pmatrix} \qquad x\begin{pmatrix} a-\lambda \\ d \\ g \end{pmatrix} + y\begin{pmatrix} b \\ e-\lambda \\ h \end{pmatrix} + z\begin{pmatrix} c \\ f \\ k-\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

Algebraic Computation

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$x\begin{pmatrix} a-\lambda \\ c \end{pmatrix} + y\begin{pmatrix} b \\ d-\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Conditions that should be satisfied by λ:

$$\frac{a-\lambda}{c} = \frac{b}{d-\lambda}$$

$$(a-\lambda)(d-\lambda) - bc = 0$$

Eigen equation

Characteristic equation

$$\lambda^2 - (a+d)\lambda + (ad-bc) = 0$$

Eigen values and Eigen vectors of a large matrix

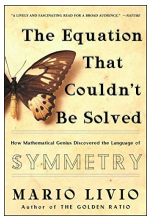
$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1n} \\ \vdots & \ddots & & & \vdots \\ a_{i1} & & a_{ii} & & a_{in} \\ \vdots & & & \ddots & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nn} \end{pmatrix}$$

Characteristic equation for $n \times n$ matrix:

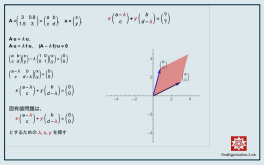
$$\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_n = 0$$

n solutions exist

But, how can we get the solutions of n[the] degree equation?



$$\mathbf{A} = \begin{pmatrix} 3 & 0.8 \\ 1.5 & 3 \end{pmatrix}$$



Eigenvalue	Eigen Vector
$\lambda_1 = 1.9,$	$\mathbf{u}_1 \propto \begin{pmatrix} -1 \\ 1.4 \end{pmatrix}$
$\lambda_2 = 4.1,$	$\mathbf{u}_2 \propto \begin{pmatrix} 1 \\ 1.4 \end{pmatrix}$

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1n} \\ \vdots & \ddots & & & \vdots \\ a_{i1} & & a_{ii} & & a_{in} \\ \vdots & & & \ddots & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nn} \end{pmatrix}, \quad \mathbf{A} \in n \times n$$

Eigenvalue	Eigen Vector
λ_1	$\mathbf{u}_1$
λ_2	$\mathbf{u}_2$
$\vdots$	$\vdots$
λ_n	$\mathbf{u}_n$

Mathematical Investigation of Leslie Matrix Model

$$\underbrace{\begin{pmatrix} n_0(t) \\ n_1(t) \\ \vdots \\ n_m(t) \end{pmatrix} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-1} b_m & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & s_{m-1} & 0 \end{pmatrix}^t \begin{pmatrix} n_0(0) \\ n_1(0) \\ \vdots \\ n_m(0) \end{pmatrix}}_{\mathbf{n}(t)=\mathbf{L}^t\mathbf{n}(0)}$$



Eigenvalues and Eigenvectors of matrix **L** ?

$$\mathbf{A} = \begin{pmatrix} a_{11} & \cdots & a_{1i} & \cdots & a_{1n} \\ \vdots & \ddots & & & \vdots \\ a_{i1} & & a_{ii} & & a_{in} \\ \vdots & & & \ddots & \vdots \\ a_{n1} & \cdots & a_{ni} & \cdots & a_{nn} \end{pmatrix}, \quad a_{ij} \geq 0 \quad \text{for all } i, j$$

$$\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \cdots + a_n = 0$$

✓The eigenvalue with the largest absolute value is always real and nonnegative.



$$\mathbf{L} = \begin{pmatrix} s_0 b_1 & s_1 b_2 & \cdots & s_{m-1} b_m & 0 \\ s_0 & 0 & \cdots & 0 & 0 \\ 0 & s_1 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & s_{m-1} & 0 \end{pmatrix}$$

$$|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_m|$$
$$\mathbf{u}_1, \quad \mathbf{u}_2, \quad \cdots, \quad \mathbf{u}_m$$

$$\begin{aligned} \mathbf{L}\mathbf{u}_1 &= \lambda_1 \mathbf{u}_1 \\ \mathbf{L}\mathbf{u}_2 &= \lambda_2 \mathbf{u}_2 \\ &\vdots \\ \mathbf{L}\mathbf{u}_m &= \lambda_m \mathbf{u}_m \end{aligned}$$

$$\begin{aligned} \begin{pmatrix} \mathbf{L}\mathbf{u}_1 & \mathbf{L}\mathbf{u}_2 & \cdots & \mathbf{L}\mathbf{u}_m \end{pmatrix} &= \begin{pmatrix} \lambda_1\mathbf{u}_1 & \lambda_2\mathbf{u}_2 & \cdots & \lambda_m\mathbf{u}_m \end{pmatrix} \\ \mathbf{L}\begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \end{pmatrix} &= \begin{pmatrix} \lambda_1\mathbf{u}_1 & \lambda_2\mathbf{u}_2 & \cdots & \lambda_m\mathbf{u}_m \end{pmatrix} \\ \mathbf{L}\begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix} &= \end{aligned}$$

$$\mathbf{L}\begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$$

$$= \begin{pmatrix} | & | & | & | \\ \lambda_1\mathbf{u}_1 & \lambda_2\mathbf{u}_2 & \lambda_3\mathbf{u}_3 & \lambda_4\mathbf{u}_4 \\ | & | & | & | \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_1\mathbf{u}_1 & \lambda_2\mathbf{u}_2 & \lambda_3\mathbf{u}_3 & \lambda_4\mathbf{u}_4 \end{pmatrix}$$



Diagonalization of matrix L

$$\mathbf{L}\begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \end{pmatrix} = \begin{pmatrix} \lambda_1\mathbf{u}_1 & \lambda_2\mathbf{u}_2 & \cdots & \lambda_m\mathbf{u}_m \end{pmatrix}$$

$$\mathbf{L}\begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix} = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$$

$$\mathbf{L}\mathbf{U} = \mathbf{U}\mathbf{\Lambda}$$

$$\mathbf{L}\mathbf{U}\mathbf{U}^{-1} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$$

$$\mathbf{U} = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix}$$

$$\mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$$



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Diagonalization of matrix L

$$\mathbf{L} \Rightarrow \mathbf{U} = \begin{pmatrix} | & | & & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_m \\ | & | & & | \end{pmatrix}$$

$$\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1} \qquad \mathbf{\Lambda} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$$

$$\mathbf{L}^t = \mathbf{U}\mathbf{\Lambda}\underbrace{\mathbf{U}^{-1}\mathbf{U}}_{\mathbf{I}}\mathbf{\Lambda}\underbrace{\mathbf{U}^{-1}\mathbf{U}}_{\mathbf{I}}\cdots\underbrace{\mathbf{U}^{-1}\mathbf{U}}_{\mathbf{I}}\mathbf{\Lambda}\mathbf{U}^{-1} = \mathbf{U}\mathbf{\Lambda}^t\mathbf{U}^{-1}$$

$$\mathbf{\Lambda}^t = \begin{pmatrix} \lambda_1^t & & 0 \\ & \ddots & \\ 0 & & \lambda_m^t \end{pmatrix}$$



Go back to the Demography

